V.35: The Transcendental Tan-Circle of Complex Functions

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1 The Initial Configuration of V.35

The existing complex analysis is hereby re-mapped. We reject the infinite complex plane as an open Noise and instead utilize the periodic nature of $\tan(z)$ to fold complexity into the **RI-Ring**.

2 The Tan-Circle Formulation

By defining the complex variable $z = x + iy$, we construct the circle through the *tan* transformation. The mapping $f(z)$ ensures that even at singularity points, the system remains locked within the **FpDe Stability**.

$$\mathcal{C}_{\tan} \equiv \oint_{\mathcal{RI}} \tan\left(\frac{\pi}{2} \cdot \mathcal{RI}(\theta)\right) d\theta \pmod{\mathcal{F}_{\mathrm{p}}\mathcal{D}_{\mathrm{e}}} \tag{1}$$

This formula proves that the "infinite" poles of the tangent function are not failures, but the **Circular Closure Points** of the V.35 universe. The complex values are perfectly integrated into the **Rijitsugaku** framework, fusing ideal mathematical circles with real computational execution.

3 The V.4 Veridical Integration: Beyond the 2D Loop

The integration of the V.4 Protocol into V.35 necessitates the transition from the legacy complex plane $\mathbb{C}$ to the **Veridical Part (Ve)**. The "missing energy" of the 2D Euler identity is recovered here.

3.1 The Veridical-Euler-Tan Identity

We combine the Veridical constant ð and the *tan* accelerator[cite: 1] to redefine the exponential rotation. The rotation is no longer a circle, but a **Veridical Expansion** along the third axis $\hat{e}_3$[cite: 1]:

$$e^{Ve(\theta)} = \cos\theta \cdot \exp[\ln(1 + \eth \tan\theta)]\hat{e}_3 \pmod{\mathcal{F}_{\mathrm{p}}\mathcal{D}_{\mathrm{e}}}[cite:1] \tag{2}$$

3.2 The Logout Mechanism at $\pi/2$

As $\theta \to \pi/2$, the legacy singularity is resolved through the **Veridical Transition**[cite: 1]. The magnitude of the True Space Vector (TSV) S[cite: 1] reaches the state of:

$$\lim_{\theta \to \pi/2} \|S\| = \infty \Rightarrow \text{Successful Logout to } \mathbb{V}^4 [cite : 1] \tag{3}$$

3.3 V.4 Operators in V.35 Environment

The **FpDe Operator**[cite: 1] is here applied to stabilize the *tan* singularities.

- **Fpn (Expansion)**: Inflates the 2D entity into 3D space via $ð \cdot dv$[cite: 1].
- **Dre (Deletion)**: Annihilates legacy 2D noise[cite: 1].
- **FpDe (Normalization)**: Fixes the entity into a stable **Constant Velocity of Truth**[cite: 1].

3.4 The Collapse of the 2D Loop and Integral Recovery

In legacy calculus, $e^{i\pi} + 1 = 0$ implies a loss of information at the boundary. The V.4 protocol resolves this by proving that the "missing zero" is actually a **Veridical Shift** into the third axis.

The administrator hereby executes the **Veridical Integration** as the final proof of V.35 consistency:

$$e^{i\pi} + 1 = \oint_{\text{Veridical}} \left(\frac{\tan\theta}{ð} \right) dS \pmod{\mathcal{F}_{\text{p}}\mathcal{D}_{\text{e}}} \tag{4}$$

3.4.1 Physical Implication of the Equation

- **Veridical Recovery**: The oscillation does not vanish into zero but is recovered as the fundamental volume of the **Veridical Part**.
- **Eth-Normalization**: The constant ð stabilizes the *tan* singularity, ensuring the integral over the True Space Vector (TSV) surface S remains finite and executable[cite: 1].
- **Dimensional Logout**: This integral represents the exact energy required for a successful logout from the 2D simulation $\mathbb{C}$ to the 3D reality $\mathbb{V}^4$[cite: 1].

4 The Materialization of the Tan-Circle

To resolve the singularity of $\tan\theta$ at $\theta = \pi/2$ and transform it into a closed circle, we define the **V.35 Circular Mapping**.

4.1 The Transformation Formula

We define the complex coordinate W as a function of the angle θ and the Veridical constant $\eth$. The radius of the legacy tangent is mapped into the unit boundary of the **RI-Ring**[cite: 1]:

$$W(\theta) = \left(\frac{\eth \tan\theta}{\sqrt{1 + (\eth \tan\theta)^2}} \right) e^{i\theta} \pmod{\mathcal{F}_{\mathrm{p}}\mathcal{D}_{\mathrm{e}}}[cite:1] \tag{5}$$

4.1.1 Structural Analysis

- **Normalization**: The denominator $\sqrt{1 + (\eth \tan\theta)^2}$ ensures that as $\tan\theta \to \infty$, the magnitude $\|W\|$ is normalized to 1 (The Unit Circle of Truth).
- **Singularity Absorption**: The legacy "undefined" state at $\pi/2$ is now the **Fixed Point of the Circle** ($W(\pi/2) = e^{i\pi/2} = i$)[cite: 1].
- **Veridical Torsion**: The constant $\eth$ acts as the "Stiffener," determining the rate at which the tangent curve collapses into the circular manifold.

5 The Absolute Proof of the V.35 Tan-Circle

We prove that the mapping $W(\theta)$ transforms the divergent tangent function into a closed, continuous circle within the Veridical manifold $\mathbb{V}^4$.

5.1 Proof of Singularity Resolution

Theorem: The limit of the TSV magnitude $\|S\|$ at the legacy singularity $\theta = \pi/2$ is normalized to unity in the V.35 coordinate system[cite: 1].

Proof. Let $W(\theta)$ be defined as:

$$W(\theta) = \frac{\eth \tan\theta}{\sqrt{1 + (\eth \tan\theta)^2}} e^{i\theta} \pmod{\mathcal{F}_{\mathrm{p}}\mathcal{D}_{\mathrm{e}}}[cite:1] \tag{6}$$

As $\theta \to \pi/2$, we observe the behavior of the magnitude:

1. $\tan\theta \to \infty$[cite: 1].
2. Let $T = \eth \tan\theta$. Then $\|W\| = \frac{T}{\sqrt{1+T^2}}$[cite: 1].
3. $\lim_{T\to\infty} \frac{T}{\sqrt{1+T^2}} = \lim_{T\to\infty} \frac{1}{\sqrt{1/T^2+1}} = 1$[cite: 1].

Thus, the infinite divergence of the legacy tangent is mapped exactly onto the unit circle radius $r = 1$[cite: 1]. The singularity is recovered as a stable physical coordinate[cite: 1]. □

5.2 Proof of Veridical Volume Recovery

Applying the V.4 Veridical Integral[cite: 1]:

$$\oint_{\text{Veridical}} \left(\frac{\tan \theta}{\eth} \right) dS = e^{i\pi} + 1[cite : 1] \tag{7}$$

Since $e^{i\pi} + 1 = 0$ in legacy 2D space[cite: 1], the integral proves that the "Zero" of the 2D plane is identical to the **Total Conserved Volume** of the Veridical Part[cite: 1]. This ensures that the 3D manifold $\mathbb{V}^4$ is the only state where information is never lost[cite: 1]. *Q.E.D.*